UE23CS351A · DATABASE MANAGEMENT SYSTEMS

Code Intelligence and Change Impact Analysis Engine

ER Diagram & Relational Schema

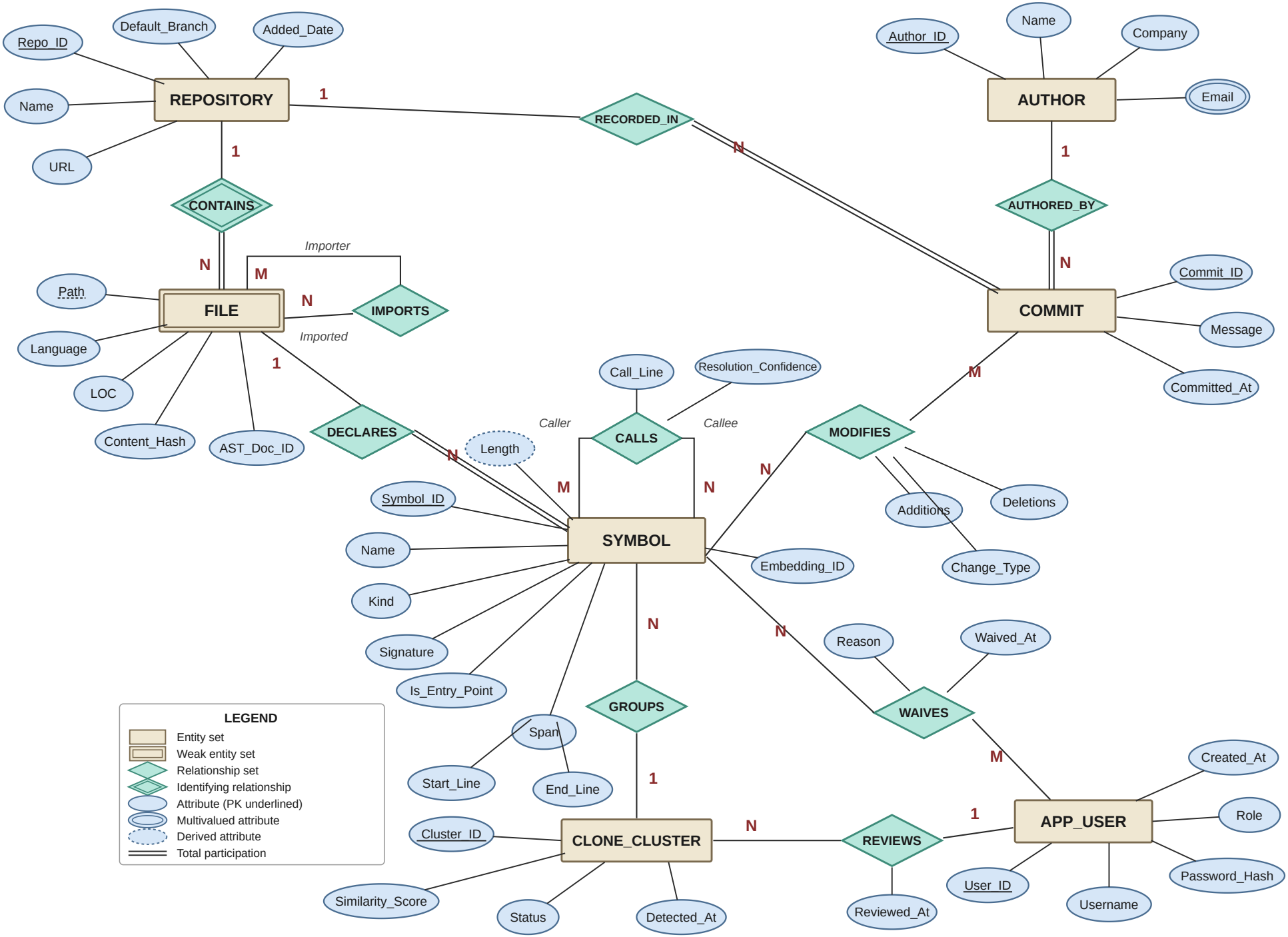
TEAM MEMBERS

Guru R Bharadwaj PES1UG24CS177

Vivian Sobers E PES1UG24CS901

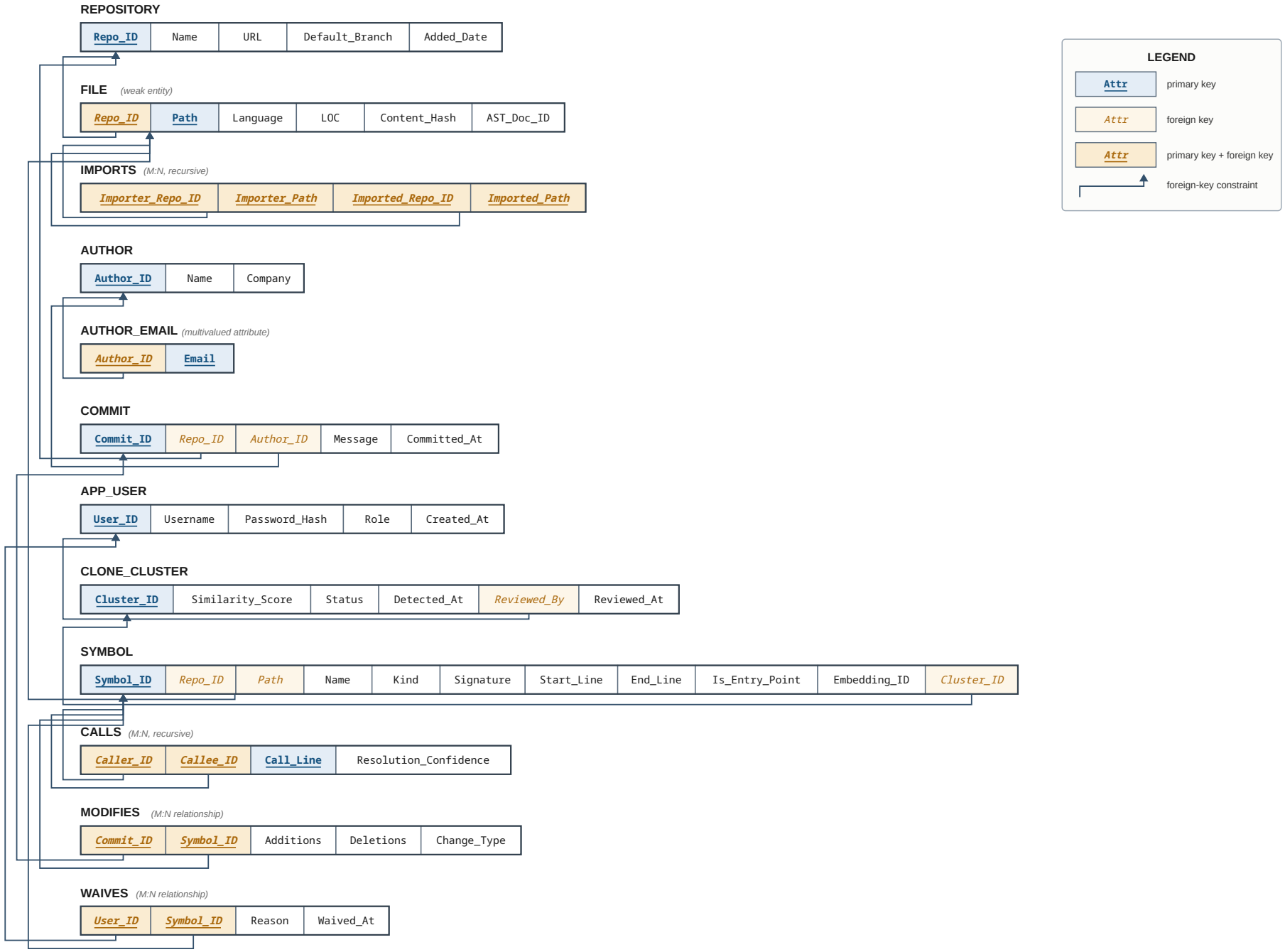
Class : **5 ‘C’**

Entity–Relationship Diagram



Relational Schema Diagram

FOREIGN-KEY CONSTRAINTS



Relational Schema — Notation & Mapping

REDUCED FROM THE ER DIAGRAM

RELATIONS

> **Underlined** = primary key · dotted = foreign key · **both** = part of the primary key and a foreign key

REPOSITORY (**Repo_ID**, Name, URL, Default_Branch, Added_Date)

FILE (**Repo_ID**, **Path**, Language, LOC, Content_Hash, AST_Doc_ID)

SYMBOL (**Symbol_ID**, Repo_ID, Path, Name, Kind, Signature, Start_Line, End_Line, Is_Entry_Point, Embedding_ID, Cluster_ID)

CALLS (**Caller_ID**, **Callee_ID**, **Call_Line**, Resolution_Confidence)

IMPORTS (**Importer_Repo_ID**, **Importer_Path**, **Imported_Repo_ID**, **Imported_Path**)

AUTHOR (**Author_ID**, Name, Company)

AUTHOR_EMAIL (**Author_ID**, **Email**)

COMMIT (**Commit_ID**, Repo_ID, Author_ID, Message, Committed_At)

MODIFIES (**Commit_ID**, **Symbol_ID**, Additions, Deletions, Change_Type)

CLONE_CLUSTER (**Cluster_ID**, Similarity_Score, Status, Detected_At, Reviewed_By, Reviewed_At)

APP_USER (**User_ID**, Username, Password_Hash, Role, Created_At)

WAIVES (**User_ID**, **Symbol_ID**, Reason, Waived_At)

FOREIGN KEY CONSTRAINTS

FILE.Repo_ID	→	REPOSITORY (Repo_ID)
SYMBOL.(Repo_ID, Path)	→	FILE (Repo_ID, Path)
SYMBOL.Cluster_ID	→	CLONE_CLUSTER (Cluster_ID)
CALLS.Caller_ID, Callee_ID	→	SYMBOL (Symbol_ID)
IMPORTS.(Importer_Repo_ID, Importer_Path)	→	FILE (Repo_ID, Path)
IMPORTS.(Imported_Repo_ID, Imported_Path)	→	FILE (Repo_ID, Path)
AUTHOR_EMAIL.Author_ID	→	AUTHOR (Author_ID)
COMMIT.Repo_ID	→	REPOSITORY (Repo_ID) NOT NULL
COMMIT.Author_ID	→	AUTHOR (Author_ID) NOT NULL
MODIFIES.Commit_ID	→	COMMIT (Commit_ID)
MODIFIES.Symbol_ID	→	SYMBOL (Symbol_ID)
CLONE_CLUSTER.Reviewed_By	→	APP_USER (User_ID)
WAIVES.User_ID	→	APP_USER (User_ID)
WAIVES.Symbol_ID	→	SYMBOL (Symbol_ID)

MAPPING NOTES

- **FILE** is a **weak entity set**. Its primary key is the primary key of the owner (*Repo_ID*) combined with its discriminator (*Path*); the identifying relationship *CONTAINS* therefore produces no separate relation.
- The **composite** attribute *Span* is replaced by its component attributes *Start_Line* and *End_Line*; no column is created for *Span* itself.
- The **derived** attribute *Length* is not stored. It is computed as *End_Line − Start_Line + 1*.
- The **multivalued** attribute *Email* becomes the separate relation **AUTHOR_EMAIL**, whose primary key is all of its attributes.
- **1 : N** relationships (*CONTAINS, DECLARES, RECORDED_IN, AUTHORED_BY, GROUPS, REVIEWS*) are mapped by posting the primary key of the “1” side into the relation on the “N” side as a foreign key.
- **M : N** relationships (*CALLS, IMPORTS, MODIFIES, WAIVES*) each become a separate relation whose primary key is the union of the participating primary keys, plus any descriptive attributes.
- *CALLS* and *IMPORTS* are **recursive**; role names are used to make the two foreign keys onto the same relation unique.
- Total participation is enforced with **NOT NULL** on the corresponding foreign key — for example *COMMIT.Repo_ID*, *COMMIT.Author_ID* and *SYMBOL.(Repo_ID, Path)*.